

AMONI EROT KEVIN

SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Software Developer with 6 years of experience building secure, high-availability backend systems and APIs for government and financial platforms. Specialized in microservices architecture, RESTful APIs, and PostgreSQL database engineering, with proven experience supporting real-time transaction systems and third-party payment integrations including M-PESA and Flutterwave. Strong background in designing scalable, secure applications with OAuth2, RBAC, and audit-compliant data handling. Experienced across full SDLC using Agile methodologies, with production deployments on Linux-based cloud environments using Docker and Kubernetes. Focused on delivering reliable, performant systems aligned with operational efficiency, data security, and financial system integrity.

CORE COMPETENCIES

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| ▶ Full-Stack Web Application Development | ▶ RESTful API Design & System Integration | ▶ Secure Coding & RBAC Authentication |
| ▶ Node.js / NestJS / TypeScript / React.js | ▶ PostgreSQL & Relational Database Design | ▶ Docker & Kubernetes Containerisation |
| ▶ GitHub Actions CI/CD Pipelines | ▶ Production Monitoring (Prometheus/Grafana) | ▶ Business Continuity & DR Support |
| ▶ Agile/Scrum & Full SDLC Participation | ▶ Cross-functional Stakeholder Collaboration | ▶ Technical Documentation & Code Reviews |

TECHNICAL SKILLS

Frontend: React.js, Next.js, TypeScript, Angular (familiar), Tailwind CSS, HTML5/CSS3 production UI systems across enterprise and government platforms.

Backend: Node.js, NestJS (primary), Express.js, Python FastAPI RESTful API development, microservice architecture, secure server-side logic.

Databases: PostgreSQL (advanced SQL, schema design, query optimisation, indexing), MySQL-compatible SQL, Redis, dbt (ELT transformation).

DevOps & Containers: Docker, Kubernetes (GKE), GitHub Actions CI/CD, Terraform (IaC), AWS (EC2, ELB, RDS, S3), GCP full production deployment and infrastructure ownership.

Security & Compliance: OAuth 2.0, JWT, RBAC, OWASP-compliant input validation, TLS/SSL, audit trail logging, access control enforcement implemented across regulated and financial platforms.

Monitoring & BCP: Prometheus (5-target: API, PostgreSQL, Redis, Node Exporter, cAdvisor), Grafana dashboards, alerting, backup/restore procedures, disaster recovery testing.

APIs & Integration: OpenAPI/Swagger contract design, webhook handling, third-party API integration (M-PESA Daraja, Flutterwave, Africa's Talking), service-oriented architecture.

PROFESSIONAL EXPERIENCE

iWorld Afric Jun 2025 – Present · Nairobi, Kenya

Full-Stack Developer

End-to-end full-stack development across multiple client-facing production platforms — NestJS/TypeScript backends, React/Next.js frontends, PostgreSQL data layers, Docker/Kubernetes deployment.

- Reduced average API response time by 45% as measured by Prometheus latency metrics, by refactoring NestJS service layers, introducing Redis caching for high-frequency queries, and optimising 12 critical PostgreSQL queries with composite indexes and query plan analysis.
- Accelerated feature delivery by 35% as measured by sprint velocity across 6 Agile cycles, by designing reusable NestJS module templates and React component libraries that eliminated redundant scaffolding across 4 concurrent client projects.
- Eliminated 100% of critical authentication vulnerabilities identified in internal security reviews, by implementing OAuth 2.0 guard-based authentication, JWT refresh token rotation, and RBAC-enforced endpoint protection across all client platforms.
- Improved data reliability to 99.8% accuracy as measured by automated validation test suites, by integrating field-level input validation (class-validator, Zod), PostgreSQL CHECK constraints, and integration test coverage across all API endpoints before production release.
- Cut deployment time from 45 minutes to under 8 minutes as measured by GitHub Actions pipeline logs, by building containerised Docker deployments with multi-stage builds and automating rollout to GCP Kubernetes Engine via CI/CD pipelines.
- Enabled 3 institutional client integrations within 2 weeks each as measured by go-live dates, by designing well-documented OpenAPI/Swagger API contracts with clear versioning, error schemas, and integration guides reducing partner onboarding friction significantly.

State Department for Diaspora Affairs

Jan 2025 – Dec 2025 · Nairobi, Kenya

Full-Stack Developer

Full production ownership for a regulated government digital platform serving 10,000+ active users — TypeScript/Node.js backend, React frontend, AWS infrastructure, strict compliance requirements.

- Improved platform uptime from 94% to 99.2% as measured by monthly availability reports over 6 months, by deploying a 5-target Prometheus + Grafana observability stack that detected and resolved 11 critical performance bottlenecks before they caused user-facing outages.
- Reduced manual data processing time by 40% as measured by operations team weekly reporting logs, by building automated Python ETL pipelines that ingested, validated, and transformed large-scale international diaspora datasets — eliminating 14 hours of manual work per week.
- Secured government platform data for 10,000+ users with zero post-deployment security incidents, by implementing OAuth 2.0 authentication, RBAC access controls, TLS encryption, and full audit trail logging in compliance with government data protection requirements.
- Cut release cycle defect rate by 60% as measured by post-release bug reports over 4 consecutive sprints, by introducing API contract testing (Postman), database migration validation, and structured SIT/UAT sign-off checklists before every production deployment.
- Decreased API integration time for 4 partner systems from an average of 3 weeks to 5 days, by designing RESTful APIs with comprehensive OpenAPI documentation, sandbox environments, and step-by-step integration guides that enabled partner developers to self-serve onboarding.
- Maintained 100% disaster recovery readiness as evidenced by 3 successful quarterly DR test exercises, by implementing automated PostgreSQL and Redis backup scheduling to AWS S3 with tested restoration procedures documented in the team's BCP runbook.

Turk Analytica

Jan 2023 – Jun 2024 · Nairobi, Kenya

Software Developer & Data Engineer

Backend development and data systems engineering for client analytics platforms across Kenya.

- Achieved 99.9% data accuracy in monthly client reporting outputs as measured by automated validation audits, by building Python data quality pipelines with systematic error detection, null-checking, and reconciliation logic across 6 client data streams.
- Reduced manual data cleaning effort by 40% as measured by team time-tracking logs, by developing reusable Python automation scripts that standardised ingestion, transformation, and validation workflows across disparate client data sources.
- Delivered Power BI dashboards that reduced management reporting time by 3 hours per week per client, by building pre-modelled datasets with DAX measures and scheduled refresh enabling self-service analytics without requiring analyst intervention.

Turkana County Government – Dept. of Economic Planning *Jun 2020 – Dec 2022 · Turkana, Kenya*

Software Developer

Software development and system automation for government data platforms supporting 500,000+ residents across Turkana County.

- Reduced weekly report generation time by 70% from 10 hours to under 3 hours as measured by operations logs by engineering automated data pipeline applications that replaced manual data extraction, transformation, and formatting workflows end-to-end.
- Influenced a KES 2 million county budget reallocation to high-impact development zones as evidenced by the 2022 County Budget Review Report, by delivering a data-driven analysis of poverty alleviation programme performance supported by 3 years of historical programme data.
- Built sustainable data management capacity across 30+ county government staff as measured by post-training assessment scores averaging 82%, by standardising data collection templates and delivering structured Excel and digital tools training over 4 workshop sessions.

KEY PROJECT DELIVERIES

Agiza – Healthcare-Analogous Patient-Facing Digital Platform · Lead Full-Stack Developer

Production · 80% reduction in service wait times · 99%+ uptime · Live payment processing · AWS/GCP/Kubernetes
agiza-blond.vercel.app

- Reduced customer wait times by 80% as measured by pre/post platform adoption surveys across 3 pilot restaurants, by building a real-time QR-based ordering system with WebSocket state synchronisation directly analogous to patient-facing clinical workflow systems in a healthcare setting.
- Achieved 99%+ platform uptime over 6 months as measured by Prometheus availability metrics, by deploying on AWS EC2/ELB + GCP GKE with Kubernetes rolling deployments, automated health checks, and GitHub Actions CI/CD pipelines with zero unplanned downtime incidents.
- Processed 100% of transactions without payment failure as measured by M-PESA Daraja callback logs over the first 90 days of production, by implementing idempotency keys, retry logic, and webhook reconciliation in the payment integration layer.

Innobid – Enterprise Document Intelligence Platform · Lead Full-Stack Developer

Live production · innobid.net · 70% efficiency improvement · Multi-institutional deployment
innobid.net

- Improved procurement processing efficiency by 70% as measured by client time-to-decision benchmarks across 3 institutional partners, by architecting a multi-agent AI platform (CrewAI, OpenAI, Claude, LangChain) with a NestJS backend and PostgreSQL data warehouse that automated document analysis and vendor evaluation workflows.
- Reduced partner API integration time from 3 weeks to 4 days as measured across 3 onboarding cycles, by designing OpenAPI/Swagger-documented RESTful APIs with sandbox environments, versioning, and self-service integration guides directly translating to third-party system integration experience relevant to healthcare ERP and HRMIS environments.

MKiba – Financial Management Platform (Regulated Domain) · Full-Stack Developer

React · FastAPI · PostgreSQL · Flutterwave · RBAC · Audit trails · Financial compliance

- Eliminated 100% of unauthorised cross-role data access incidents as measured by post-launch security testing, by implementing granular RBAC controls, JWT authentication, and comprehensive audit trail logging demonstrating secure coding practices applicable to regulated healthcare system environments.
- Maintained zero financial reconciliation errors across 6 months of production as measured by monthly reconciliation reports, by building PostgreSQL transaction schemas with atomic operations, constraint enforcement, and automated balance validation checks that prevented data inconsistencies before they reached reporting layers.

EDUCATION

Moi University

2018 – 2022 · Eldoret, Kenya

BSc. Applied Statistics with Computing

Computing systems, software engineering, database design, statistical analysis, and applied mathematics — providing strong analytical foundations for translating complex clinical and operational requirements into technical solutions.

Holberton School – ALX Program
Software Engineering – Full Stack

2023 – 2024 · Remote

Systems programming, backend engineering, databases, Linux/Unix, DevOps, API design, security, and cloud deployment — with emphasis on production-grade engineering standards.

ALX Africa
Data Science Fellow — ML, Advanced Statistics, Cloud (AWS)

2023 – 2024 · Remote

REFERENCES

Available upon request.